READER PATHWAY / TECHNICAL

Independent engineering evaluation

Behavioural and evidence-oriented architecture for reviewers without releasing BTG proprietary source or developer manuals.

Edition 2.0 · 23 October 2026 · Educational/public explanatory documentation · No public investment offering or approval claimed

Engineering evaluation without publishing implementation

BTG intentionally keeps the seven proprietary systems' source, unpublished algorithms, secrets, private implementation contracts and build chains closed. Independent engineering understanding is still possible at the **system-boundary** level: define inputs and outputs, constraints, failure behaviour, trustworthy evidence and operational limits. A reviewer requiring source-level assurance must use a controlled evaluation process under a suitable agreement; a public explanatory guide is not a replacement.

Component-by-component contracts to evaluate

System	Public behavioural contract	Crucial failure test
Bridge	Authorize source, purpose, fields and recipient; deliver only allowed outputs	Revoked/expired scope is rejected and reported
NIKI	Frozen candidate proposal, independent challenge, exact failure retention	Candidate with incorrect holdout cannot be promoted
FarmNodes	Bounded task on identified host with isolated access and evidence	Cross-host claim fails unless genuinely separate physical hosts tested
Circuits	Typed components connected under permitted version and rights	Invalid unit/type mismatch halts or rejects

System	Public behavioural contract	Crucial failure test
BTDU	Authoritative payload and permitted semantic reconstruction	Reconstruction hash mismatch or unauthorized view rejected
ADAM	Policy-governed state transition with evidence	Expired grant cannot alter protected state
Exchange	Asset / issuer / instrument separation; audited market and contract events	Ledger simulation cannot be treated as legal issuance or bank settlement
ENTITY (separate public component)	Identity, provenance and signed authority references	Signature alone cannot create legal title

Acceptance stages an independent evaluator should recognize

1. Describe the frozen artifact and allowed input/output domain.
2. Record environment, operating-system/tool versions, hashes and authorization scope.
3. Test normal, invalid, adversarial and boundary conditions, preserving actual failures.
4. Demonstrate independent oracle or reference comparison on withheld tasks.
5. Verify authority, rights and privacy limits separately from computation.
6. Demonstrate genuine host and operator separation when claiming independent distributed execution.
7. Report safety, latency and resource failures with measured bounds, not marketing language.
8. Verify rollback, denial and incident reporting; classify any missing evidence as pending.

What an independent engineering reviewer must receive privately

- Architecture/data-flow diagrams showing trust and source-of-truth boundaries.
- Signed exact-version artifacts, reproducible run instructions and test harness within an authorized scope.
- Test vectors and holdouts selected by the reviewer or an independent evaluator.
- Evidence of actual host enrollment, mutual authentication and access control.
- Controlled source review or black-box inspection under a protected evaluation agreement.
- Performance baselines with equivalent quality, hardware, power, memory and wall-time metrics.

Do not confuse numerical counts

5,126 registered type definitions or 2,310 logical bindings do not prove thousands of physical laboratories. A result of 65,536/65,536 in one bounded algorithm does not establish autonomous invention on unrelated tasks. A local passing test is not a production acceptance test or independent third-party certification.